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Abstract

This coursework examine classical cryptography with a focus on the vigenere cipher. It explains basic encryption and decryption concepts and reviews the origin and working process of the vigenere algorithm. Several weaknesses of the traditional Vigenere cipher are identified, such as key repetition, limited character support, and vulnerability to cryptanalysis. To address these issues, a modified and improved Vigenere algorithm is proposed using ASCII values and the XOR operation. This hybrid approach allows the use of alphabets, numbers, and special character, increasing randomness and security. The proposed algorithm is tested with multiple examples, and the results show successful encryption and decryption. Overall, the modified algorithm provides better confidentiality and integrity compared to the traditional Vigenere cipher, while remaining simple to implement and understand.

The report begins by introduction the fundamental concepts of cryptography and its role in securing information. It explains the basic principles of encryption and decryption and outlines the difference between symmetric and asymmetric encryption techniques. A brief historical overview includes to show how cryptography evolved from early substitution ciphers to more structured algorithms.

The Vigenere cipher is then analysed in detail, covering its background, mathematical model and encryption and decryption procedures. Although it was historically considered secure, the coursework highlights several weaknesses, including its repeating key structure, restriction to uppercase alphabetic characters, and susceptibility to cryptanalytic attacks such as the kasiski examination and frequency analysis.

To overcome these limitations, a modified Vigenere algorithm is proposed. This improved approach combines ASCII encoding with the XOR logical operation, allowing encryption at the binary level and supporting a wider range of characters. The use of XOR increase randomness and reduces predictable patterns in the ciphertext and the proposed algorithm is tested using multiple test cases. The results confirm accurate encryption and decryption while increasing more complex ciphertext than the classical Vigenere cipher.

1. Introduction:



Figure 1: Cryptography (LinkedIn.com)

The science of **using mathematics** to **encrypt** and **decrypt** data is known as cryptography (PACHGHARE, 2019). It refers to the **art of secret writing**. Using various cryptographic techniques, the user can secure their message. These methods allow him or her to transmit or store the message safely. It offers **security** concepts like **non-repudiation, confidentiality, integrity** and **authentication** (PACHGHARE, 2019).

It includes various number of methods, including **digital signatures**, **hashing**, and **cryptographic protocols**. Blockchain secure **communications** and **authentication** systems all make use of it. Even if data is **intercepted**, cryptography **guarantees** that it **cannot be deciphered** without the **right key**. Additionally, it helps companies in safeguarding for maintain confidential data against cyberattacks and illegal access (PACHGHARE, 2019).

1.1 Origin and Background:

HISTORY OF CRYPTOGRAPHY

digicert®

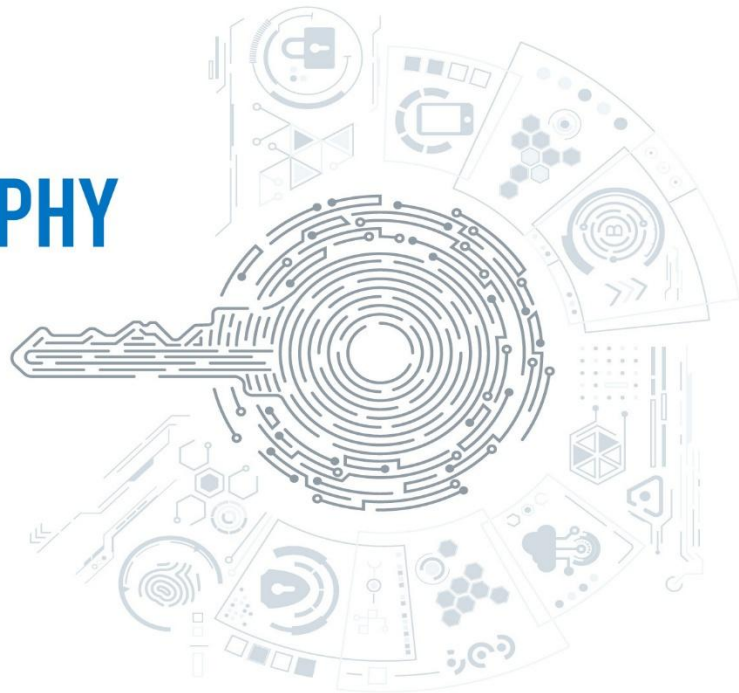


Figure 2: History of Cryptography (Digicert.com)

The ancient era of cryptography was primarily concerned with message meaning concealment while in transit. The first type of cryptography was **secret writing**, which was practiced due to the **lack of reading skills** among most people. The **Egyptian scribes** practiced the earliest type of cryptography by using **hieroglyphs** that were written in a **non-standard way** so that the message was not interpreted by unintended individuals as early as **1900 B.C.** The **scytale method** was later **developed** by the **Greeks**, who wrote the message down on a **rolled cloth around a rod**. The **Romans** later improved cryptography by **developing the Caesar Shift Cipher**, which is a type of monoalphabetic letter substitute (Damico, 2009).

During the **middle age** of time in history, cryptography **evolved** and **grew** in sophistication as the flaws within existing cryptography were perfected. A major milestone occurred with the **development of frequency** analysis by **Arab cryptographers** in 1000 C.E., which enabled existing monoalphabetic encryption to be cracked. In light of

deficiencies in cryptography, there came the **polyalphabetic substitution** method by **Leon Battista Alberti**, also regarded as **the “Father of Western Cryptology”** which significantly enhanced security by involving a **combination of substitution alphabets** rather than **only one** (Damico, 2009).

In today's world, the field of cryptography has moved ahead to include not only the secrecy of messages but also **integrity verification**, **authentication**, and **digital signatures**. **Gilbert Vernam** enhanced the earlier methods of ciphers by **inventing** the **Vernam-Vigenère cipher** in 1918, which increased the **level of security** in the encryption methods. Cryptography was also applied in practical applications, like in the case of the **"Navajo Code Talkers"** that used their language to transmit messages covertly. In modern days, the use of **public key** and **private key** cryptography has become **extremely common** and has now become the basis of **digital signatures** and **secure communications**. There are many safer and secure cryptography have been introduced in modern world some of them are **AES, DES, IDES, RSA** and **Elliptic curve Cryptography** (Damico, 2009).

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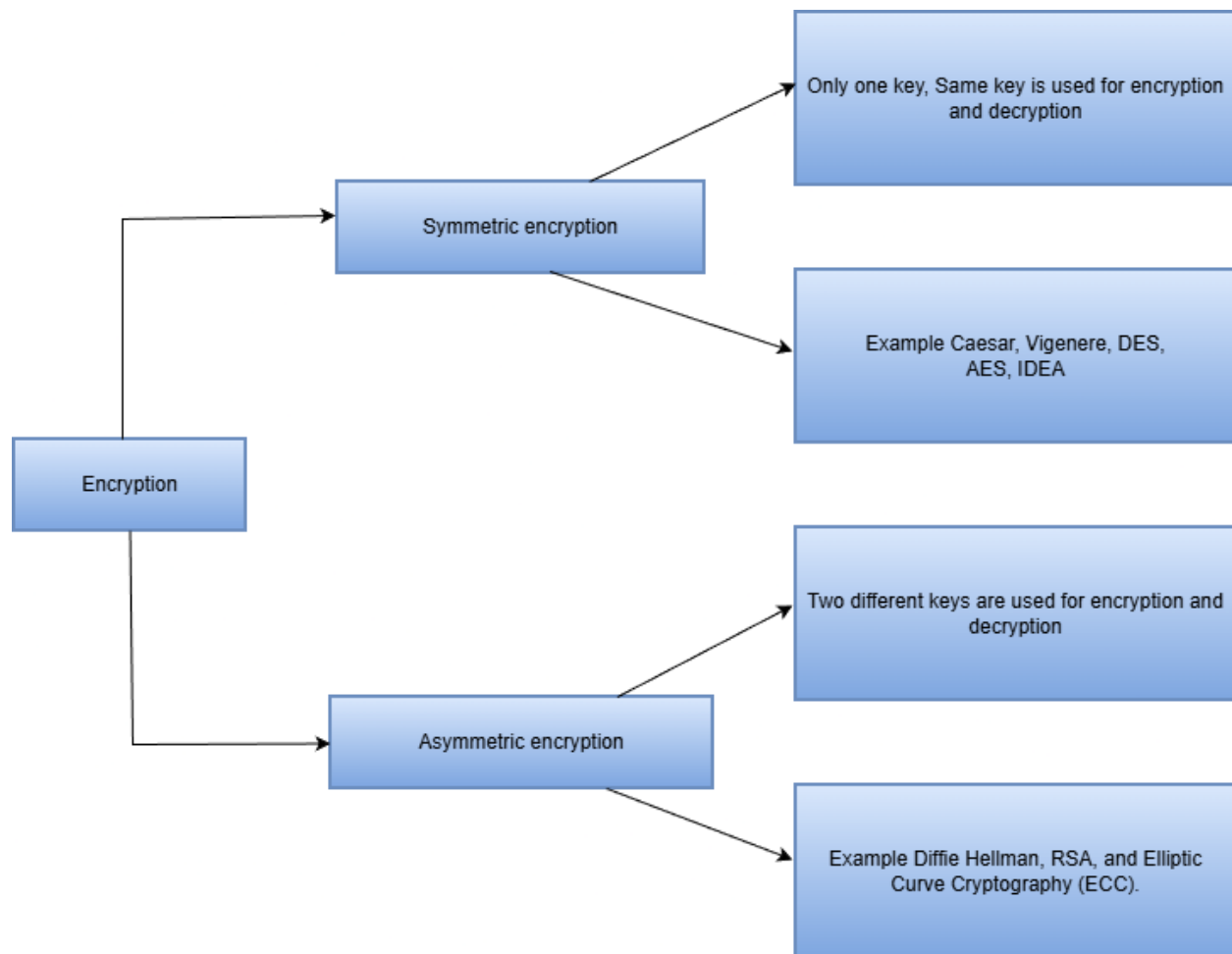


Figure 3: Encryption Technique (Draw.io.com)

1.2 Encryption:

The process of converting **plaintext data** into **ciphertext** with a **secret code** is known as encryption. A **secret key** or **keys** is used to accomplish **encrypting process**. We can increase **data security** by using **encryption**. There are two kinds of traditional encryption methods, depending on how many keys are used for encryption and decryption (PACHGHARE, 2019).

1.2.1 Symmetric encryption: (Using single key for encryption and decryption)

In this method, only **one key is needed** for symmetric encryption. The data is encrypted and decrypted using the **same key**. Various symmetric encryption algorithms exist such as **Caesar, Vigenere, DES, AES, IDEA**, and **3DES** are a few of them (PACHGHARE, 2019).

1.2.2 Asymmetric encryption: (Also called public key encryption)

In this method, two distinct keys are needed for asymmetric encryption. There is a mathematical relationship between these keys. These keys are referred to as **public** and **private keys**. A public key is one that is accessible to **everyone**, while a private key is one that **only receiver** has to decrypt the **ciphertext**. Asymmetric encryption algorithms are numerous like **Elliptic Curve Cryptography (ECC)**, **RSA**, and **Diffie-Hellman** etc. (PACHGHARE, 2019).

1.3 Decryption:

The process of **converting ciphertext** into the **original plaintext** is known as **decryption**. Decryption requires a **secret key**. A **secret key** or **keys** is used to accomplish **decrypting process**. It guarantees that **data integrity** has not been **compromised during transmission**. Decryption produces **inaccurate results** if the data has been **tampered**. By verifying whether the correct key **successfully decrypts** the data, it verifies that it originated from a **reliable sender** (PACHGHARE, 2019).

2. Aim and Objectives:

- The Aim of this coursework is to find, develop and test cryptography algorithm by building upon an existing cipher, while focusing on to enhance data security by ensuring confidentiality and integrity against potential threats.

The Objective of this coursework focuses one specific cryptography method while understanding how encryption and decryption process works in the cryptography. The more detailed objective is listed below:

- ❖ To research on existing cryptography and select a base cipher.
- ❖ To understand how vigenere algorithm works.
- ❖ To identify a weakness in the Vigenere algorithm.
- ❖ To mitigate existing vulnerabilities of the Vigenere algorithm.
- ❖ To self-modified and improve Vigenere Algorithm.
- ❖ To use the ASCII values table in the encryption and decryption process.
- ❖ To use the XOR Operation for making encryption process strong.
- ❖ To test encryption and decryption of algorithm using multiple test cases.
- ❖ To maintain confidentiality and integrity against adversaries.
- ❖ To discuss advantages and disadvantages of developed cryptography.
- ❖ To discuss applications areas of the cryptography.

3. Analysis of Specific Cryptography Algorithm:

3.1 Origin and Background:



Figure 4: Blaise de Vigenere (wikipedia.org)

Vigenere cipher was originally described by **Giovan Pietro Battista Bellaso** in his book **La cifra del Sig. Giovan Battista Bellaso**, published in 1553. In this, he improved upon Trithemius "tabula recta," but included a "countersign" or "key" that would switch from letter to letter. Alberti and Trithemius' systems were fixed pattern of substitution, a pattern of substitutions in Bellaso's method could easily be modified by merely picking a new key. Another similar but more powerful autokey cipher was described before the court of Henry III of France by **Blaise de Vigenere** in 1586. Later on in the 19th century, the creation of the Bellaso cipher was wrongly attributed to **Vigenere** (babylon, 2013).

In the book titled "The Codebreakers" by David Kahn. The Codebreakers lamented that it is unfortunate that this contribution was misplaced, as history ignored this important contribution and instead named a regressive and elementary cipher for him Vigenère, even though he had nothing to do with it (babylon, 2013).

3.2 Vigenere Cipher:

Vigenere cipher is a way of encrypting alphabetic texts using a sequence of different Caesar cipher Shifts, **each based on the letter of a keyword**. Vigenere cipher is a polyalphabetic substitution cipher that forms **a 26 by 26 matrix** of the shifts in the Caesar cipher. It comprises a group of monoalphabetic substitution cipher rules based on the shifts of the Caesar cipher that are **0 through 25**. The Vigenere cipher was deemed safe for over 200 years but was later shown to be insecure in 1917. However, its weakness was found much later. **Friedrich Kasiski** was able to discover how to find the period and thus the key and the plaintext. The central underlying principle of Vigenere cipher is that it aims at avoiding simple frequency analysis attacks on plaintext letter frequencies. However, the most important weakness in the principle of Vigenere cipher is that its key repeats itself. In case the cryptanalyst manages to figure out the length of the key, then the cipher can be broken very easily. There are numerous tools like Kasiski test and Friedman test that can be used in identifying the length of the cipher's key. However, the Vigenere cipher can be broken much easily because of the availability of computers (Aized Amin Soofi, 2016).

Most cipher text can be decrypted in a few seconds even for a long key. Now this cipher is known as a **trivial problem** and does **not ensure** any **level of security** as far as current standards are concerned. Vigenere encryption is performed by adding each index of the plain text letter with the index of the password letter as represented in the Vigenere square or Vigenere square table as shown in figure 4 (Aized Amin Soofi, 2016).

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Figure 5: Vigenere Square Table (CTF Handbook.com)

The process of encryption and decryption of a vigenere algorithm is written below mathematically (Aliyu, 2016).

$C_i = E(P_i + K_i) \bmod 26$ (Encryption Formula)

$P_i = D(C_i - K_i) \bmod 26$ (Decryption Formula)

Where C_i is the ciphertext.

P_i is the plaintext character.

K_i is the key.

E is the encryption, while D is the decryption.

3.3 Encryption Process of Vigenere Algorithm:

Encryption is a process of transforming information, **known as plaintext** using an algorithm to make it unreadable producing what is **known as ciphertext**, except to those who possess special information, usually referred to as a key. It remains the most efficient method of achieving data security (Mohammed, 2021).

Let's suppose, we have a plaintext: **"Cyber Security"**, and a keyword: **"Future"**. The encryption and decryption process of vigenere algorithm is done and shown below:

Plaintext: **"CYBER SECURITY"**

Keyword: **"FUTUREFUTUREF"**

Encryption Formula: $C_i = E(P_i + K_i) \bmod 26$

$C = E(P_i + K_i) \bmod 26 : E(C + F) \bmod 26 : E(2 + 5) \bmod 26 : 7 : "H"$
$Y = E(P_i + K_i) \bmod 26 : E(Y + U) \bmod 26 : E(24 + 20) \bmod 26 : 44 - 25 : 19 : "T"$
$B = E(P_i + K_i) \bmod 26 : E(B + T) \bmod 26 : E(1 + 19) \bmod 26 : 20 : "U"$
$E = E(P_i + K_i) \bmod 26 : E(E + U) \bmod 26 : E(4 + 20) \bmod 26 : 24 : "Y"$
$R = E(P_i + K_i) \bmod 26 : E(R + R) \bmod 26 : E(17 + 17) \bmod 26 : 34 - 25 : 9 : "J"$
$S = E(P_i + K_i) \bmod 26 : E(S + E) \bmod 26 : E(18 + 4) \bmod 26 : 22 : "W"$
$E = E(P_i + K_i) \bmod 26 : E(E + F) \bmod 26 : E(4 + 5) \bmod 26 : 9 : "J"$
$C = E(P_i + K_i) \bmod 26 : E(C + U) \bmod 26 : E(2 + 20) \bmod 26 : 22 : "W"$
$U = E(P_i + K_i) \bmod 26 : E(U + T) \bmod 26 : E(20 + 19) \bmod 26 : 39 - 25 : 14 : "O"$
$R = E(P_i + K_i) \bmod 26 : E(R + U) \bmod 26 : E(17 + 20) \bmod 26 : 37 - 25 : 12 : "M"$
$I = E(P_i + K_i) \bmod 26 : E(I + R) \bmod 26 : E(8 + 17) \bmod 26 : 25 : "Z"$
$T = E(P_i + K_i) \bmod 26 : E(T + E) \bmod 26 : E(19 + 4) \bmod 26 : 23 : "X"$
$Y = E(P_i + K_i) \bmod 26 : E(Y + F) \bmod 26 : E(24 + 5) \bmod 26 : 29 - 25 : 4 : "E"$

Table 1: Vigenere Encryption Process

After encrypting the plaintext: **"Cyber Security"** with keyword: **"FutureFutureF"**. Finally, we got Vigenere Ciphertext: **"HTUYJWJWOMZXW"**

3.4 Decryption Process of Vigenere Algorithm:

Decryption is a process where **you take ciphertext and turn it back into genuine text** and into text that you or **the computer can read and make sense**. Decryption is used to unencode the information with keys or with the algorithm. (Mohammed, 2021).

Ciphertext: **"HTUYJWJWOMZXW"**

Keyword: **"FUTUREFUTUREF"**

Decryption Formula: $P_i = D(C_i - K_i) \bmod 26$

$H = D(C_i - K_i) \bmod 26 : D(H - F) \bmod 26 : D(7 - 5) \bmod 26 : 2 : "C"$
$T = D(C_i - K_i) \bmod 26 : D(T - U) \bmod 26 : D(19 - 20 + 25) \bmod 26 : 44 - 20 : 24 : "Y"$
$U = D(C_i - K_i) \bmod 26 : D(U + T) \bmod 26 : D(20 - 19) \bmod 26 : 1 : "B"$
$Y = D(C_i - K_i) \bmod 26 : D(Y - U) \bmod 26 : D(24 - 20) \bmod 26 : 4 : "E"$
$J = D(C_i - K_i) \bmod 26 : D(J - R) \bmod 26 : D(9 - 17 + 25) \bmod 26 : 34 - 17 : 17 : "R"$
$W = D(C_i - K_i) \bmod 26 : D(W - E) \bmod 26 : D(22 - 4) \bmod 26 : 18 : "S"$
$J = D(C_i - K_i) \bmod 26 : D(J - F) \bmod 26 : D(9 - 5) \bmod 26 : 4 : "E"$
$W = D(C_i - K_i) \bmod 26 : D(W - U) \bmod 26 : D(22 - 20) \bmod 26 : 2 : "C"$
$O = D(C_i - K_i) \bmod 26 : D(O - T) \bmod 26 : D(14 - 19 + 25) \bmod 26 : 39 - 19 : 20 : "U"$
$M = D(C_i - K_i) \bmod 26 : D(M - U) \bmod 26 : D(12 - 20 + 25) \bmod 26 : 37 - 20 : 17 : "R"$
$Z = D(C_i - K_i) \bmod 26 : D(Z - R) \bmod 26 : D(25 - 17) \bmod 26 : 8 : "I"$
$Y = D(C_i - K_i) \bmod 26 : D(Y - E) \bmod 26 : D(23 - 4) \bmod 26 : 19 : "T"$
$E = D(C_i - K_i) \bmod 26 : D(E - F) \bmod 26 : E(4 - 5 + 25) \bmod 26 : 29 - 5 : 24 : "Y"$

Table 2: Vigenere Decryption Process

Therefore, after decrypting the vigenere Ciphertext: **"HtuyjWjwomzxe"** with keyword: **"FutureFutureF"**. We got a plaintext: **"CYBER SECURITY"**.

3.5 Weakness Identification of the Vigenere Cipher:

a) Repeating Nature of its keyword:

Repetitive nature of the key is the major weakness that gives rise to other weaknesses. As it **repetitions**, it is relatively far **easier** to estimate the **length of the key**. The length of the key can be found out far quicker compared to **brute force techniques** and using the **Friedman** and the **Kasiski examination**. Instead, a running key cipher could be used, which is basically a Vigenere cipher but where the key is longer than the message, typically a passage from a book or something similar, or a onetime pad (tutorialspoints.com, 2025).

b) Vulnerable to Kasiski examination:

The Kasiski examination, also known as the **Kasiski test** or **Kasiski method**, is a foundational cryptanalytic technique developed to **determine the length** of the **repeating keyword** in polyalphabetic substitution ciphers, particularly the Vigenere cipher, by identifying **repeated sequences of letters** such as **digrams** or **trigrams** in the ciphertext and calculating the distances between their occurrences (Rubinstein-Salzedo, 2018).

c) No uses of Symbol or Special character:

The classical Vigenere cipher operates only on alphabetic text (**A - Z**). It's does not include support for **symbol, numbers, special characters**, which makes more **vulnerable** to the algorithm and easy to **decode** the ciphertext. It is not suitable for modern plaintext that includes symbols, numbers and Special character which enhance security level in the data. (wikipedia.org, 2025).

d) Short keys and Predictable:

It can be predictable if the key is shorter **than the plaintext** so **repeats keywords**, that **create patterns form** in the encrypted text that **reveal clues** about the key and **plaintext structure**. There are many crypto analyst methods such as Friedrich Kasiski test and **Friedman test** which can actually predict **the keyword** and so, easily **decode** the actual message of the sender. (Geek, 2019).

e) Limited Uppercase Alphabetic Letters:

The classical vigenere algorithm support only **uppercase letters** to encrypt plaintext which **increase risk level** to easily **decode** the ciphertext of this algorithm. The traditional algorithm like vigenere is not safer and protective than modern algorithm like AES, RSA and SDES. There is always chance of keyword been **predictable** or **been brute force attack** and always **vulnerable** from cryptanalysis methods such as **Friedrich Kasiski** and **the Friedman test** (Christensen, 2015).

f) Similar to Caesar Cipher:

The Caesar Cipher is a particular **instance of the Vigenere Cipher**, also known as a one, constant shift. The point of difference in Caesar Cipher is a particular example of a **mono-alphabetic letter substitution** cipher which makes use of a single, **fixed letter shift** while encoding messages. The Vigenere cipher is a more complex, poly-alphabetic letter substitution cipher which **alters letter shifts** according to a **particular keyword** and effectively composed of multiple **Caesar ciphers** using shifts **ranging from 0 to 25** (Kumar, 2025).

4. Development Phase of Vigenere Algorithm:

4.1 Improvement Strategies for the Vigenere Cipher:

a) Adding Symbols, Special character or Numbers in keywords:

The classical vigenere algorithm is limited to alphabet character (A - Z) only, which make much vulnerable comparing to modern cryptography algorithm. **Now adding Symbols like ! @ # \$ % * - , + _ = < >**, Special characters **like À ü ë ì ß ñ** and Numbers **like from 0 – 9 in the keyword**, the ciphertext becomes **more significantly stronger**. This expansion in the keyword makes much harder for attacker to predict actual key and brute force the keyword and reduces vulnerability from cryptanalysis methods such as Friedrich Kasiski and the Friedman test.

b) Use of ASCII Values Alphabets Table:

The vigenere algorithm only supports uppercase alphabetic latters (A-Z) and cipher which constitutes **a matrix of 26 by 26** Caesar cipher shifts. It limits the ability to encrypt numbers, symbols and special characters. Adding the **ASCII values of character**, the cipher can be extended to handle a printable character, **including lowercase latter, digit, punctuation, and symbols with new alphabetic values**.

c) Use of XOR Operator for Encryption and Decryption:

To enhance or make much stronger vigenere cipher, the use **of XOR operator** for encryption and decryption is **perfect combination with ASCII values table**. In other hand XOR operator **is reversible** in the same applied **key, easy, simple and fast** for encrypting and decrypting text. It also **reduces** the chance been keyword **predictable** and **brute force attack**. It is more secure than traditional vigenere cipher technique.

4.2 Modified and Improved Vigenere Algorithm:

4.2.1 Hybrid ASCII-XOR Vigenere Algorithm:

The classical vigenere cipher is also known polyalphabetic encryption technique however, this algorithm suffers from different security vulnerabilities such as key repetition, vulnerability to cryptanalysis, predictable and limited to uppercase alphabets only so, the way to overcome this weakness is **hybrid ASCII-XOR algorithm**. The ASCII values algorithm **support encryption of numbers, symbol and special characters**, while the XOR operation provides a **simple reversible mechanism** for both **encryption** and **decryption**. This hybrid approach of ASCII and XOR help algorithm to **enhance security**, increase **keyword strength** from been broken **by brute force** or **any cryptanalysis methods**.

ASCII stands for **American Standard Code for Information Interchange** refers to the standard character coding in telecommunication (geeksforgeeks, 2025). It refers to a standard for character encoding, which assigns numbers to letters, digits, special marks, and other characters. The use of ASCII code helps to represent the alphanumeric data (geeksforgeeks, 2025).

The encryption and decryption process of the modified Vigenere cipher are can be represented by mathematically:

$C_i = E(P_i \text{ XOR Keyword})$ (Encryption Formula)

$P_i = D(C_i \text{ XOR Keyword})$ (Decryption Formula)

Where C_i is the ciphertext.

P_i is the plaintext character.

K_i is the key.

XOR is the XOR Operation.

E is the encryption, while D is the decryption.

4.2.2 Encryption Process of Modified Vigenere Algorithm:

One of the important roles in encrypting plaintext is **played** by the **ASCII table**, also known as the **American Standard Code for information interchange**. It is used to represent characters in the numeric form, allowing plaintext to be converted from **binary to decimal** and **decimal to hexa-decimal** for successfully **encrypting** plaintext to modified cipher text.

The ASCII code

American Standard Code for Information Interchange

www.theasciicode.com.ar

ASCII control characters				ASCII printable characters												Extended ASCII characters											
DEC	HEX	Simbolo ASCII		DEC	HEX	Simbolo	DEC	HEX	Simbolo	DEC	HEX	Simbolo	DEC	HEX	Simbolo	DEC	HEX	Simbolo	DEC	HEX	Simbolo	DEC	HEX	Simbolo	DEC	HEX	Simbolo
00	00h	NULL (carácter nulo)		32	20h	espacio	64	40h	@	96	60h	`	128	80h	Ç	160	A0h	á	192	C0h	Ł	224	E0h	Ó			
01	01h	SOH (inicio encabezado)		33	21h	!	65	41h	A	97	61h	a	129	81h	ü	161	A1h	í	193	C1h	ł	225	E1h	ô			
02	02h	STX (inicio texto)		34	22h	"	66	42h	B	98	62h	b	130	82h	é	162	A2h	ó	194	C2h	Ł	226	E2h	Ô			
03	03h	ETX (fin de texto)		35	23h	#	67	43h	C	99	63h	c	131	83h	â	163	A3h	ú	195	C3h	ł	227	E3h	Û			
04	04h	EOT (fin transmisión)		36	24h	\$	68	44h	D	100	64h	d	132	84h	ä	164	A4h	ñ	196	C4h	ł	228	E4h	ô			
05	05h	ENQ (enquiry)		37	25h	%	69	45h	E	101	65h	e	133	85h	å	165	A5h	Ñ	197	C5h	ł	229	E5h	Õ			
06	06h	ACK (acknowledgement)		38	26h	&	70	46h	F	102	66h	f	134	86h	ä	166	A6h	°	198	C6h	ł	230	E6h	µ			
07	07h	BEL (timbre)		39	27h	'	71	47h	G	103	67h	g	135	87h	ç	167	A7h	°	199	C7h	ł	231	E7h	¶			
08	08h	BS (retroceso)		40	28h	(	72	48h	H	104	68h	h	136	88h	ê	168	A8h	¿	200	C8h	ł	232	E8h	ß			
09	09h	HT (tab horizontal)		41	29h	)	73	49h	I	105	69h	i	137	89h	ë	169	A9h	®	201	C9h	ł	233	E9h	Ü			
10	0Ah	LF (salto de línea)		42	2Ah	*	74	4Ah	J	106	6Ah	j	138	8Ah	è	170	AAh	™	202	CAh	ł	234	EAh	Ù			
11	0Bh	VT (tab vertical)		43	2Bh	+	75	4Bh	K	107	6Bh	k	139	8Bh	ï	171	ABh	½	203	CBh	ł	235	EBh	Ú			
12	0Ch	FF (form feed)		44	2Ch	,	76	4Ch	L	108	6Ch	l	140	8Ch	ì	172	ACH	¼	204	CCh	ł	236	ECh	Ý			
13	0Dh	CR (retorno de carro)		45	2Dh	.	77	4Dh	M	109	6Dh	m	141	8Dh	í	173	ADh	½	205	CDh	ł	237	EDh	Û			
14	0Eh	SO (shift Out)		46	2Eh	.	78	4Eh	N	110	6Eh	n	142	8Eh	Ā	174	ACh	«	206	CEh	ł	238	EEh	Ŷ			
15	0Fh	SI (shift In)		47	2Fh	/	79	4Fh	O	111	6Fh	o	143	8Fh	Ā	175	AFh	»	207	CFh	ł	239	EFh	ŷ			
16	10h	DLE (data link escape)		48	30h	0	80	50h	P	112	70h	p	144	90h	Ē	176	B0h	ˆ	208	D0h	ł	240	FOh				
17	11h	DC1 (device control 1)		49	31h	1	81	51h	Q	113	71h	q	145	91h	æ	177	B1h	ˆ	209	D1h	ł	241	F1h	±			
18	12h	DC2 (device control 2)		50	32h	2	82	52h	R	114	72h	r	146	92h	Æ	178	B2h	ˆ	210	D2h	ł	242	F2h	ˆ			
19	13h	DC3 (device control 3)		51	33h	3	83	53h	S	115	73h	s	147	93h	ø	179	B3h	ˆ	211	D3h	ł	243	F3h	ˆ			
20	14h	DC4 (device control 4)		52	34h	4	84	54h	T	116	74h	t	148	94h	ö	180	B4h	ˆ	212	D4h	ł	244	F4h	ˆ			
21	15h	NAK (negative acknowle.)		53	35h	5	85	55h	U	117	75h	u	149	95h	ö	181	B5h	ˆ	213	D5h	ł	245	F5h	ˆ			
22	16h	SYN (synchronous idle)		54	36h	6	86	56h	V	118	76h	v	150	96h	ù	182	B6h	ˆ	214	D6h	ł	246	F6h	ˆ			
23	17h	ETB (end of trans. block)		55	37h	7	87	57h	W	119	77h	w	151	97h	ù	183	B7h	ˆ	215	D7h	ł	247	F7h	ˆ			
24	18h	CAN (cancel)		56	38h	8	88	58h	X	120	78h	x	152	98h	ÿ	184	B8h	ˆ	216	D8h	ł	248	F8h	ˆ			
25	19h	EM (end of medium)		57	39h	9	89	59h	Y	121	79h	y	153	99h	Ŏ	185	B9h	ˆ	217	D9h	ł	249	F9h	ˆ			
26	1Ah	SUB (substitute)		58	3Ah	:	90	5Ah	Z	122	7Ah	z	154	9Ah	Ŭ	186	BAh	ˆ	218	DAh	ł	250	FAh	ˆ			
27	1Bh	ESC (escape)		59	3Bh	;	91	5Bh	[	123	7Bh	{	155	9Bh	ø	187	BBh	ˆ	219	DBh	ł	251	FBh	ˆ			
28	1Ch	FS (file separator)		60	3Ch	<	92	5Ch	\	124	7Ch		156	9Ch	£	188	BCh	ˆ	220	DCh	ł	252	FCh	ˆ			
29	1Dh	GS (group separator)		61	3Dh	=	93	5Dh	]	125	7Dh	}	157	9Dh	Ø	189	BDh	ˆ	221	DDh	ł	253	FDh	ˆ			
30	1Eh	RS (record separator)		62	3Eh	>	94	5Eh	^	126	7Eh	~	158	9Eh	x	190	BEh	ˆ	222	DEh	ł	254	FEh	ˆ			
31	1Fh	US (unit separator)		63	3Fh	?	95	5Fh	-				159	9Fh	f	191	BFh	ˆ	223	DFh	ł	255	FFh	ˆ			
127	20h	DEL (delete)																									

Figure 6: ASCII Values Table (theasciicode.com.)

Let's take an example if we have a plaintext: **"Cyber Security"** and keyword: **"S3cure@#!1607"**. now vigenere cipher process is done by modified encryption formula.

Plaintext: **"Cyber Security"**

Keyword: **"S3cure#!607S3c"**

Converting plaintext into Binary:

Plaintext	ASCII	Binary
C	67	01000011
y	121	01111001
b	98	01100010
e	101	01100101
r	114	01100101
space	32	00100000
S	83	01010011
e	101	01100101
c	99	01100011
u	117	01110101
r	114	01100101
i	105	01101001
t	116	01110100
y	121	01111001

Converting keyword into Binary:

Keyword	ASCII	Binary
S	83	01010011
3	51	00110011
c	99	01100011
u	117	01110101
r	114	01100101
e	101	01100101
#	35	00100011
!	33	00100001
6	54	00110110
0	48	00110000
7	55	00110111
S	83	01010011
3	51	00110011
c	99	01100011

Table 3: ASCII and Binary for Encryption

$C_i = E(P_i \text{ XOR Key})$ (Encryption Formula)

Now, Applying XOR operation on first character of plaintext C and Keyword S.

C =	01000011
S =	01010011
XOR =	00010000

XOR is 00010000.

Changing XOR binary to hexadecimal.

00010000 = 16 decimal, and 10 hexa-decimals in ASCII table.

Finally, we get our first cipher 10.

Again, Applying XOR operation remaining character and keyword.

y =	01111001
3 =	00110011
XOR =	01001010

XOR is 01001010.

Changing XOR binary to hexadecimal

01001010 = 74 decimal and 4A hexadecimal in ASCII table.

b =	01100010
c =	01100011
XOR =	00000001

XOR is 00000001.

Changing XOR binary to hexadecimal

00000001 = 1 decimal and 01 hexadecimal in the ASCII table.

e =	01100101
u =	01110101
XOR	00010000

XOR is 00010000.

Changing XOR binary to hexadecimal

00010000 = 16 decimal and 10 hexadecimal in the ASCII table.

r =	01110010
r =	01110010
XOR =	00000000

XOR is 00000000.

Changing XOR binary to hexadecimal

00000000 = 0 decimal and 00 hexadecimal in the ASCII table.

Plaintext	Keyword	XOR	Decimal	Hexadecimal
Space = 00100000	e = 01100101	01000101	69	45
S = 01010011	# = 00100011	01110000	112	70
e = 01100101	! = 00100001	01000100	68	44
c = 01100011	6 = 00110110	01010101	85	55
u = 01110101	0 = 00110000	01000101	69	45
r = 01110010	7 = 00110111	01000101	69	45
i = 01101001	S = 01010011	00111010	58	3A
t = 01110100	3 = 00110011	01000111	71	47
y = 01111001	c = 01100011	00011010	26	1A

Table 4: Modified Vigenere Encryption Process

After encrypting with modified Vigenere algorithm plaintext: **“Cyber Security”** with keyword: **“S3cure#!607S3c”** finally we got our Vigenere Ciphertext **“10 4A 01 10 00 45 70 44 55 45 45 3A 47 1A”**

4.2.3 Decryption Process of Modified Vigenere Algorithm:

Ciphertext: “**10 4A 01 10 00 45 70 44 55 45 45 3A 47 1A**”

Keyword: “**S3cure#!607S3c**”

$P_i = D(C_i \text{ XOR Key})$ (Decryption Formula)

Cipher	Keyword	XOR	Decimal	Plaintext
00010000	S = 01010011	01000011	67	C
01001010	3 = 00110011	01111001	121	y
00000001	c = 01100011	01100010	98	b
00010000	u = 01110101	01100101	101	e
00000000	r = 01110010	01110010	114	r
01000101	e = 01100101	00100000	32	Space
01110000	# = 00100011	01010011	83	S
01000100	! = 00100001	01100101	101	e
01010101	6 = 00110110	01100011	99	c
01000101	0 = 00110000	01110101	117	u
01000101	7 = 00110111	01110010	114	r
00111010	S = 01010011	01101001	105	i
01000111	3 = 00110011	01110100	116	t
00011010	c = 01100011	01111001	121	y

Table 5: Modified Vigenere Decryption Process

Therefore, after decrypting modified Vigenere Ciphertext: “**10 4A 01 10 00 45 70 44 55 45 45 3A 47 1A**” with Keyword: “**S3cure#!607S3c**”. we finally get our plaintext: “**Cyber Security**”.

4.3 Modified and Improved Vigenere Algorithm Flowchart:

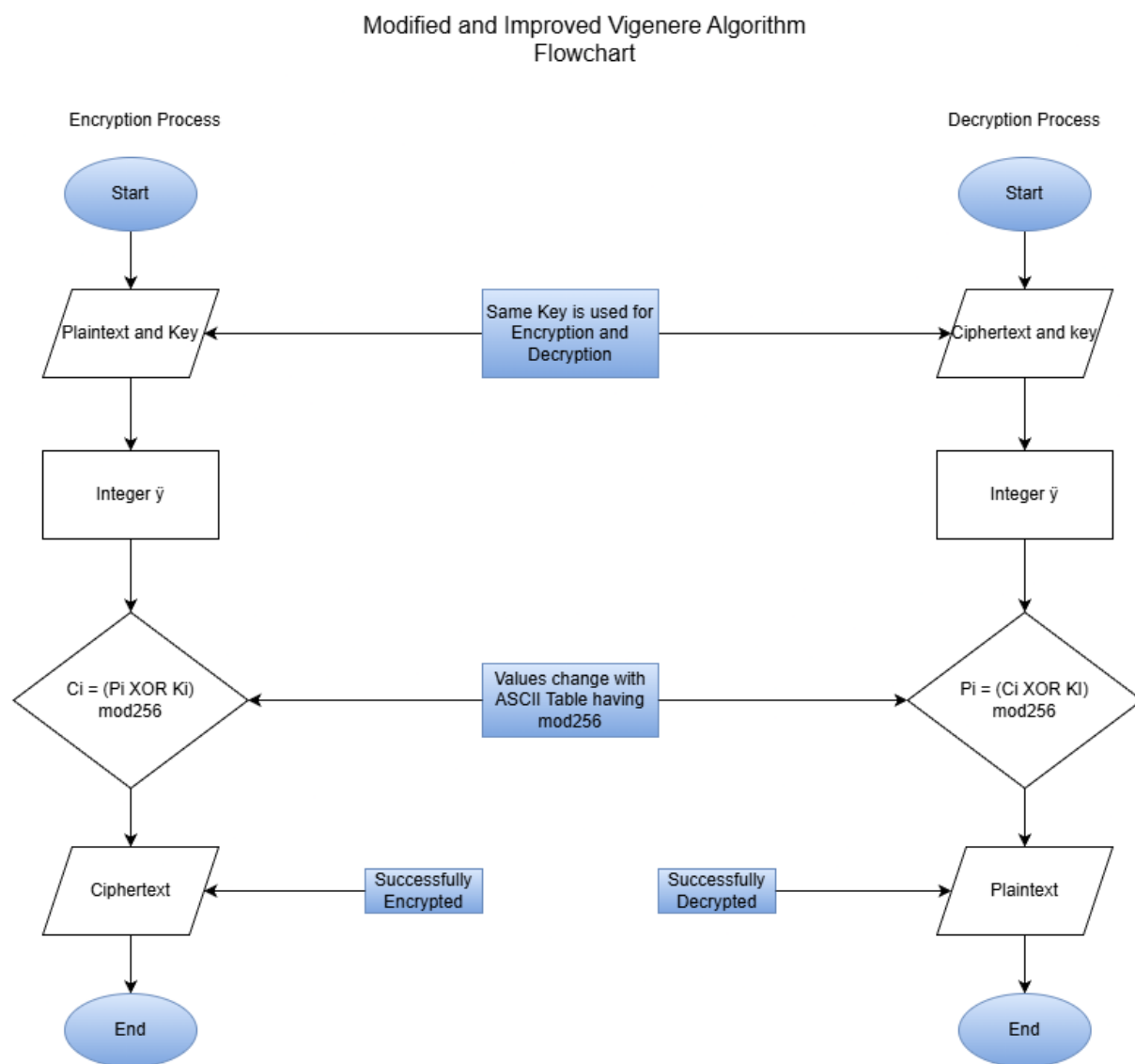


Figure 7: Modified Vigenere Flowchart (Draw.io)

5. Testing:

1. Let's suppose, we have a plaintext: **“Computing”** and keyword: **“Cyb3r&123”**. Here, encryption and decryption process is done by modified vigenere algorithm formula.

Encryption Process:

Plaintext	Keyword	XOR	Decimal	Hexadecimal
C = 01000011	C = 01000011	00000000	0	00
o = 01101111	y = 01111001	00010110	22	16
m = 01101101	b = 01100010	00001111	15	0F
p = 01110000	3 = 00110011	01000011	67	43
u = 01110101	r = 01110010	00000111	7	07
t = 01110100	& = 00100110	01010010	82	52
i = 01101001	1 = 00110001	01011000	88	58
n = 01101110	2 = 00110010	01011100	92	5C
g = 01100111	3 = 00110011	01010100	84	54

Table 6: Encryption Testing

Modified Vigenere Ciphertext: **“00 16 0F 43 07 52 58 5C 54”**

Decryption Process:

Keyword	Ciphertext	XOR	Decimal	Plaintext
C = 01000011	00000000	01000011	67	C
y = 01111001	00010110	01101111	111	o
b = 01100010	00001111	01101101	109	m
3 = 00110011	01000011	01110000	112	p
r = 01110010	00000111	01110101	117	u
& = 00100110	01010010	01110100	116	t
1 = 00110001	01011000	01101001	105	i
2 = 00110010	01011100	01101110	110	n
3 = 00110011	01010100	01100111	103	g

Table 7: Decryption Testing

Plaintext: **“Computing”**

Algorithm test	1
Plaintext	“Computing”
Ciphertext	“00 16 0F 43 07 52 58 5C 54”
Decrypt text	“Computing”
Conclusion	Pass

Table 8: Test no 1 Successful

2. Again, Let's take another example we have plaintext: **“Eñcryption”** and keyword: **“L0ck#!0L0c”**. All the encryption and decryption process is done through modified Vigenere algorithm formula.

CI = E(Pi XOR Keyword) (Encryption)

Encryption process:

Plaintext	Keyword	XOR	Decimal	Hexadecimal
E = 01000101	L = 01001100	00001001	9	09
ñ = 11110001	0 = 00110000	11000001	193	C1
c = 01100011	c = 01100011	00000000	0	00
r = 01110010	k = 01101011	00011001	25	19
y = 01111001	# = 00100011	01011010	90	5A
p = 01110000	! = 00100001	01010001	81	51
t = 01110100	0 = 00110000	01000100	68	44
i = 01101001	L = 01001100	00100101	37	25
o = 01101111	0 = 00110000	01011111	95	5F
n = 01101110	c = 01100011	00001101	13	D

Modified Vigenere Ciphertext: **“09 C1 00 19 5A 51 44 25 5F D”**

Pi = D(Ci XOR Keyword) (Decryption)

Decryption Process:

Keyword	Ciphertext	XOR	Decimal	Plaintext
L = 01001100	00001001	01000101	69	E
0 = 00110000	11000001	11110001	193	ñ
c = 01100011	00000000	01100011	99	c
k = 01101011	00011001	01110010	114	r
# = 00100011	01011010	01111001	121	y
! = 00100001	01010001	01110000	112	p
0 = 00110000	01000100	01110100	116	t
L = 01001100	00100101	01101001	105	i
0 = 00110000	01011111	01101111	111	o
c = 01100011	00001101	01101110	110	n

Plaintext: **“Eñcryption”**

Algorithm test	2
Plaintext	"Eñcryption"
Ciphertext	"09 5E 00 19 5A 51 44 25 5F D"
Decrypt text	"Eñcryption"
Conclusion	Pass

Table 9: Test no 2 Successful

3. Let's suppose, we have again Plaintext: **“Sëcüre”** and keyword: **“Shield”**. All the encryption and decryption are done through modified and improved vigenere algorithm formula.

CI = E(Pi XOR Ki) (Encryption Formula)

Encryption Process:

Plaintext	Keyword	XOR	Decimal	Hexadecimal
S = 01010011	S = 01010011	00000000	0	00
ë = 11101011	h = 01101000	10000011	131	83
c = 01100011	i = 01101001	00001010	10	0A
ü = 11111100	e = 01100101	10011001	153	99
r = 01110010	l = 01101100	00011110	30	1E
e = 01100101	d = 01100100	00000001	1	01

Modified Vigenere Ciphertext: **“00 83 0A 99 1E 01”**

Pi = D(Ci XOR Keyword) (Decryption)

Decryption Process:

Keyword	Ciphertext	XOR	Decimal	Plaintext
S = 01010011	00000000	01010011	83	S
h = 01101000	10000011	11101011	235	ë
i = 01101001	00001010	01100011	99	c
e = 01100101	10011001	11111100	252	ü
l = 01101100	00011110	01110010	114	r
d = 01100100	00000001	01100101	101	e

Plaintext: **“Sëcüre”**

Algorithm test	3
Plaintext	“Sëcüre”
Cipher text	“00 83 0A 99 1E 01”
Decrypt text	“Sëcüre”
Conclusion	Pass

Table 10: Test no 3 Successful

4. Let's suppose, we have again Plaintext: **"Crÿptò"** and keyword: **"Sÿstêm"**. All the encryption and decryption are done through modified and improved vigenere algorithm formula.

CI = E(Pi XOR Ki) (Encryption)

Encryption Process:

Plaintext	Keyword	XOR	Decimal	Hexadecimal
C = 01000011	S = 01010011	00010000	16	10
r = 01110010	ÿ = 11111111	10001101	141	8D
ÿ = 11111111	s = 01110011	10001100	140	8C
p = 01110000	t = 01110100	00000100	4	04
t = 01110100	ê = 11101010	10011110	158	9E
ò = 11010011	m = 01101101	10111110	190	BE

Modified Vigenere Ciphertext: **"10 8D 8C 04 9E BE"**

PI = D(Ci XOR Ki) (Decryption)

Keyword	Ciphertext	XOR	Decimal	Plaintext
S = 01010011	00010000	01000011	67	C
ÿ = 11111111	10001101	01110010	114	r
s = 01110011	10001100	11111111	255	ÿ
t = 01110100	00000100	01110000	112	p
ê = 11101010	10011110	01110100	116	t
m = 01101101	10111110	11010011	242	ò

Plaintext: **"Crÿptò"**

Algorithm test	4
Plaintext	"Crÿptò"
Cipher text	"10 8D 8C 04 9E BE"
Decrypt text	"Crÿptò"
Conclusion	Pass

Table 11: Test no 4 Successful

5. Let's suppose, we have again Plaintext: **"Plaintext"** and keyword: **"Keyword@1"**. All the encryption and decryption are done through modified and improved vigenere algorithm formula.

Ci = E(PI XOR Ki) (Encryption)

Encryption Process:

Plaintext	Keyword	XOR	Decimal	Hexadecimal
P = 01010000	K = 01001011	00011011	27	1B
l = 01101100	e = 01100101	00001001	9	09
a = 01100001	y = 01111001	00011000	24	18
i = 01101001	w = 01110111	00011110	30	1E
n = 01101110	o = 01101111	00000001	1	01
t = 01110100	r = 01110010	00000110	6	06
e = 01100101	d = 01100100	00000001	1	01
x = 01111000	@ = 01000000	00111000	56	38
t = 01110100	1 = 01100001	00010101	21	15

Modified Vigenere Ciphertext: **"1B 09 18 1E 01 06 01 38 15"**

Pi = D(Ci XOR Ki) (Decryption)

Decryption Process:

Keyword	Ciphertext	XOR	Decimal	Plaintext
K = 01001011	00011011	01010000	80	P
e = 01100101	00001001	01101100	108	l
y = 01111001	00011000	01100001	97	a
w = 01110111	00011110	01101001	105	i
o = 01101111	00000001	01101110	110	n
r = 01110010	00000110	01110100	116	t
d = 01100100	00000001	01100101	101	e
@ = 01000000	00111000	01111000	120	x
1 = 01100001	00010101	01110100	116	t

Plaintext: **"Plaintext"**

Algorithm test	5
Plaintext	"Plaintext"
Cipher text	"1B 09 18 1E 01 06 01 38 15"
Decrypt text	"Plaintext"
Conclusion	Pass

Table 12: Test no 5 Successful

6. Evaluation:

Vigenere is the substitution type of cipher used in encrypting data, where in the original message structure is partially hidden in the encoded message by using a combination of several different mono-alphabetic substitution ciphers, rather than doing it with one substitution alone (Simmons, 2025).

Except that, improved and modified vigenere developed algorithm enhance more security than the traditional vigenere algorithm. Modified Vigenere algorithm includes Symbols, Special Characters and numbers. It ensures more confidentiality and integrity in data only trusted individual or person having keyword can decrypt it. Some of the advantage as well as disadvantage points of an improved and modified Algorithm are listed below:

Advantages of improved and Modified Vigenere algorithm.

- ❖ Modified Vigenere algorithm supports alphabets, numbers, symbol and special character.
- ❖ Modified Vigenere algorithm key also includes now digits, symbols and numbers.
- ❖ Modified Vigenere algorithm is based on ASCII values table having more than 256-character codes.
- ❖ Modified Vigenere algorithm also include XOR operation for enhancing security of encryption as well as decryption.
- ❖ Modified Vigenere algorithm works on increasing randomness number of encrypting ciphertext.
- ❖ Modified Vigenere algorithm is harder to be predictable or vulnerable to brute force attack.

Disadvantages of improved and Modified Vigenere algorithm.

- ❖ Modified Vigenere algorithm still done in symmetric key.
- ❖ Modified Vigenere algorithm encrypting process is done through ASCII values table, maybe its vulnerable to modern cryptanalytic.
- ❖ Modified Vigenere algorithm still have keywords repetition problem.
- ❖ Modified Vigenere algorithm is not safer than other algorithm like AES, RSA and SDES.

7. Conclusion:

This coursework focuses on finding, developing, and testing a cryptography algorithm by extending an existing cipher, and it aims at improving the security of data by considering the aspects of securing the confidentiality and integrity of the information. Throughout this coursework, an extensive study of cryptography was performed, beginning from the basics and concepts of the history of cryptography, the concepts for encrypting and decrypting, and various methods of cryptography algorithms. The classic Vigenere cipher algorithm was identified as the base algorithm, and its operation procedure was extensively analysed.

Many weaknesses of the traditional Vigenere algorithm were identified out, including key repetition, the need to apply or the algorithm supporting only to uppercase letters of the alphabet, similar to Caesar cipher, no use of any digit, symbols, special character and the ability to perform cryptanalysis attacks or brute force attack in the algorithm. Using this weakness as guidance, methods of enhancing the algorithm were performed and mitigation is carried out. The algorithm uses number, symbols, special characters to the keyword for enhancing security of the encryption process as well as for decryption process.

The main role plays use of ASCII values table which include more than 0 to 31, 127, 255-character codes for having unique values and XOR logic operation for encrypting plaintext to ciphertext. XOR logic operator was easy, fast and reversible for encryption and decryption. All the encryption and decryption process has been done in the binary form to improve security level and to erase any repeating pattern behind it.

After this, the new and enhanced algorithm for the Vigenere cipher was developed and tested using several test examples. The test output proved that the new algorithm can be used to encrypt as well as decrypt any given data while generating a complex pattern in the ciphertext compared to the original Vigenere cipher algorithm. The coursework was able to meet its aim and objective in carrying out research, development, testing, and evaluation on a modified cryptographic algorithm.

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